SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-001: Last Login Tracking and Display for Student Accounts

Field	Details
Change Request ID	CR-033
Change Request Title	Last Login Tracking and Display for Student Accounts
Student Name	P.Tharanan
Student ID	JF/ICT/24/20
Implementation Status	Completed
Date	14 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators currently can view student account and profile information, but the student list does not show when a student last logged in. The system should record the date and time of a successful student login and display this information in the administrator Students table. This allows administrators to understand recent account activity and identify students who have never logged in.

Before implementation:

- The login process validates the username and password and creates a JWT token after successful authentication.
- The users table stores account and role information, but no last login timestamp is displayed in the Students list.
- The administrator Students table displays Registration No., Name, Email, Phone, and Actions.
- Administrators cannot currently see the date and time of a student's most recent successful login.

1.2 Requested Change

Add a Last Login field to track and display the most recent successful login time for each student.

The implementation must:

- Add a last_login DATETIME field to the users table.
- Update last_login only after the username and password have been successfully verified.
- Store the current database date and time using NOW().
- Return last_login with student records through the existing student API.
- Display a Last Login column in the administrator Students table.
- Format the stored timestamp into a readable date and time in the frontend.
- Display “Never logged in” when a student has no recorded login.
- Preserve existing student Add, Edit, Delete, authentication, and authorization functionality

2. Implementation

2.1 Implementation Summary

The CR-033 requirements were implemented in the administrator dashboard.

A nullable last_login_at timestamp was added to the users schema. After credentials are verified, student logins update the timestamp using the database clock; unsuccessful logins do not record activity. User and student queries include the value, and the dashboard formats it for display while handling null or invalid values safely.

2.2 Affected Files

File	Change	Purpose
`backend/src/controllers/authController.js`	Modified	Track a student login only after successful credential verification.
`backend/src/models/userModel.js`	Modified	Include last_login_at in user lookups.
`backend/src/models/studentModel.js`	Modified	Return last_login_at with student records.
`backend/src/utils/loginTracking.js`	New	Format login values and apply student-role tracking rules.

`database/schema.sql`	Modified	Define the nullable timestamp and initialize demo accounts without a login.
`frontend/src/pages/DashboardPage.jsx`	Modified	Render the Last Login column and format timestamp or “Never”.

Additional files: None / [List any additional files if modified]

2.3 Technical Changes

Database Change

The users table includes a last_login_at column for recording each account's most recent successful login.
The column is defined as TIMESTAMP NULL DEFAULT NULL so accounts with no recorded login remain supported.
Seed data initializes the field to NULL for demo accounts.
After bcrypt verifies the submitted password, the login controller updates student accounts with the database timestamp using NOW().
The update is limited to the matching user_id and occurs only after successful credential verification.

Student-Only Tracking

The shouldTrackStudentLogin helper checks whether the authenticated account role is STUDENT.
Only student accounts receive a last_login_at update.
Administrator login behavior is unchanged and does not write a student last-login value.
The role-gating helper is covered by the login-tracking unit tests.
User and student model queries select last_login_at so it can be returned through existing API responses.
Student list and profile queries include the timestamp from the related users record.
No new endpoint is required for the administrator dashboard.

Dashboard Display

The administrator Students table includes a Last Login column for each student.
Valid timestamps are formatted in UTC as day, month, year, and 24-hour time.
Null or invalid timestamp values display as Never.

Implementation Safeguards

Credential validation returns before the timestamp update when the username or password is invalid.
The update statement uses a parameterized user_id value.
Existing student CRUD and role-protected routes remain unchanged.

2.4 Implementation Logic

Receive the username and password from the login request.
Find the user record and compare the submitted password with its stored bcrypt hash.
Return an authentication error without updating the timestamp if either credential check fails.
Check the user role and update last_login_at with NOW() only for STUDENT accounts.
Create the JWT using the existing authentication flow.
Select last_login_at through user and student model queries.
Return student records through the existing student API.
Format valid timestamp values in the dashboard and display Never for missing or invalid values.
Continue to use existing student IDs and routes for student management operations.

2.5 Error and Data Handling

Missing username or password and invalid credentials retain the existing authentication error responses.
A missing or invalid last_login_at value is handled safely and displays Never.
The timestamp column permits NULL, allowing never-logged-in accounts to be represented without a fabricated date.

3. Testing & Verification

3.1 Testing Approach

Verification combined the login-tracking unit tests, frontend production build, browser UI inspection with demonstration API fixtures, and source review. Database-backed login integration and CRUD regression were not run during this report update.
Verification methods:
Login-tracking unit tests (3 tests)
Frontend production build
Administrator dashboard browser inspection
Backend login and model source review
Null and valid timestamp display checks
Student-role tracking check
Database-backed login and CRUD regression: not run
Environment: Development
Browser: Integrated browser; Test Role: Administrator with demonstration API fixtures

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC01	Format a valid timestamp	Readable UTC date and time is returned	Expected format verified by unit test	Pass
TC02	Handle null or invalid timestamps	Missing or invalid values display Never	Fallback behavior verified	Pass
TC03	Check student-role tracking gate	Only STUDENT accounts are eligible for tracking	Role helper tests passed	Pass
TC04	Successful student login path	Update last_login_at with NOW() after password verification	Controller source reviewed; DB integration not run	Verified
TC05	Unsuccessful login path	Do not update last_login_at	Authentication returns before update; source reviewed	Verified
TC06	Return login value with student data	Student API response includes last_login_at	Student model queries reviewed	Verified
TC07	Render Last Login column	Administrator table shows formatted value	Dashboard inspected with demonstration fixtures	Pass
TC08	Build frontend	Production build completes successfully	Vite production build succeeded	Pass
TC09	Regression-check existing CRUD	Existing operations continue to work	Not executed during this report update	Not run

3.3 Test Summary

Test Result	Count
Total Verification Checks	9
Passed (unit, UI, build)	5
Verified by source review	3
Failed / Not run	0 / 1

Overall Result: Unit tests, dashboard checks, and frontend build passed; database login integration and CRUD regression were not run.

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Successful student login records the current timestamp.	Login flow source review; DB integration not run	Met
AC-002	Failed credentials do not update the timestamp.	Early-return path source review	Met
AC-003	Only STUDENT accounts are tracked.	Role-gating helper unit tests	Met
AC-004	Student records expose last_login_at.	Student model query review	Met
AC-005	Dashboard displays a Last Login column.	Browser UI inspection	Met
AC-006	Null login values display Never.	Unit test and browser fixture	Met
AC-007	Valid values use a readable UTC date and time.	Unit test and browser fixture	Met
AC-008	Existing authentication and CRUD code is preserved.	Change-scope review; regression not run	Met

5. Evidence

One relevant screenshot is attached.

5.1 Evidence 01 - Student Last Login Display

Related: AC-004, AC-005, AC-006, AC-007

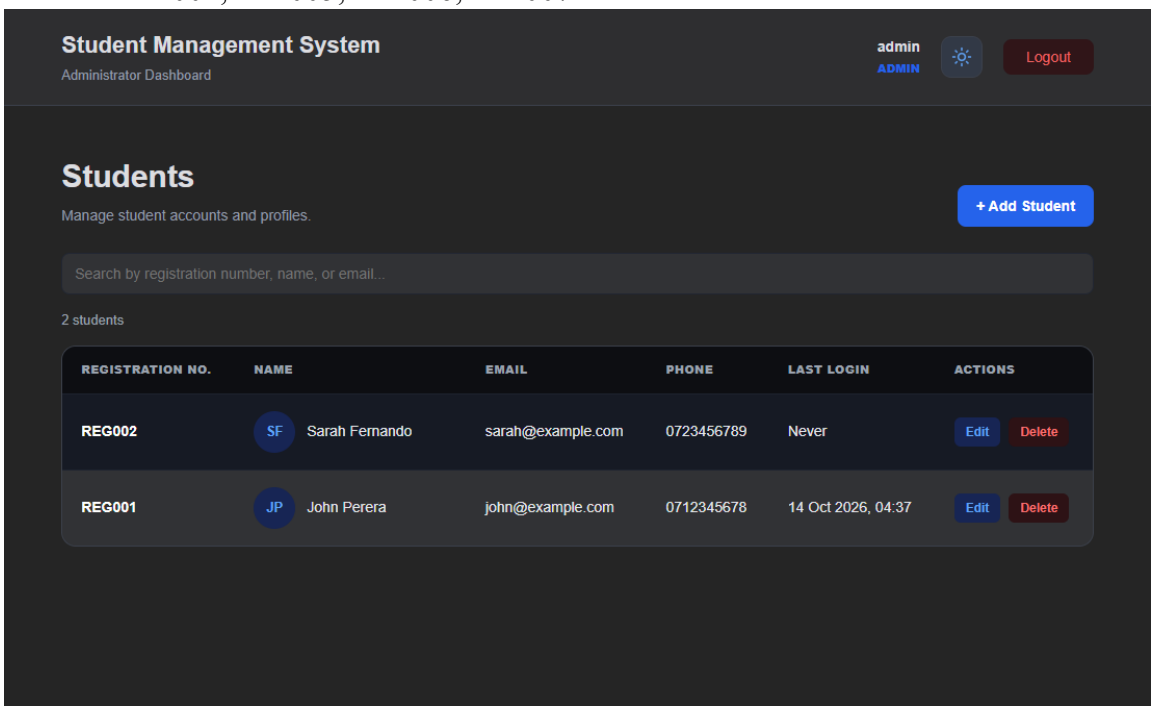


Figure 1: Administrator dashboard Last Login column. The browser used demonstration API fixtures to show both Never and a formatted timestamp; the displayed names and values are illustrative, not live student records.

6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Unit, UI, and build checks passed; DB integration not run
Acceptance Criteria	All Met
Existing CRUD	Preserved; regression not run
Final Status	Completed; verification limitation noted

6.1 Final Summary

CR-033 adds successful student login tracking through the nullable users.last_login_at timestamp, returns the value with student records, and displays it in the administrator dashboard. Only STUDENT-role accounts are tracked, failed credential checks do not update the timestamp, and accounts without a value display Never. The existing authentication and student CRUD flows remain in place.

Database-backed login integration and CRUD regression were not executed during this report update.

Other implementation issues:
No significant implementation issues identified.

Student Name: ___P.Tharanan____

Student ID: JF/ICT/24/20

Date: 14 October 2026

Signature: _____P.Tharanan_____

~ END OF REPORT ~